

Shruthi-CK / LLM-Recommender-System

<> Code

Issues

Pull requests

Actions

Projects

Security

Insights

Watch 0

This project builds a smart LLM recommender from 2,440 Reddit posts. It uses BERTopic for topic modeling, RoBERTa for sentiment, and NER to find LLMs. A knowledge base aggregates sentiment by topic/LLM, so the final app can semantically match a user query and recommend the best LLM for that task.

0 stars

0 forks

0 watching

Branches

Activity

Tags

Public repository

1 Branch

0 Tags

Go to file

t

Go to file

Add file +

Code

Shruthi-CK

Add files via upload

9c8e2f0 · 3 months ago

Knowledge Base.ipynb	Add files via upload	3 months ago
LLM Recommender.ipynb	Add files via upload	3 months ago
NER.ipynb	Add files via upload	3 months ago
README.md	Update README.md	3 months ago
Sentiment Analysis.ipynb	Add files via upload	3 months ago
Topic Modeling.ipynb	Add files via upload	3 months ago

README

LLM-Recommender-System

This project builds an LLM recommender system from scratch using Reddit data. The goal is to advise a user on the best LLM (like ChatGPT, Claude, or Gemini) for a specific task by analyzing real user sentiment for that exact use case.

The system ingests 2,440 raw Reddit posts and processes them through a multi-stage data pipeline to create a smart, semantic search application.

Pipeline & Methodology

The project is broken down into five distinct stages, each represented by a Jupyter Notebook:

1. **Topic Modeling.ipynb** :

- Ingests all 2,440 Reddit posts.
- Uses **BERTopic** to cluster the documents into 10 distinct topics, such as "Use Case: Coding & Development" and "Pain Point: Content Filters."

2. **Sentiment Analysis.ipynb** :

- Employs a **RoBERTa transformer model**, which is highly effective for social media text.
- Assigns a `positive`, `negative`, or `neutral` sentiment score to each post.

3. **NER.ipynb (Named Entity Recognition)**:

- Identifies which LLM is being discussed in each post.
- Cleverly uses regex patterns to find explicit mentions (e.g., "Grok", "GPT-4o").
- For posts without explicit mentions, it falls back to the post's subreddit (e.g., a post in `r/ClaudeAI` is tagged as "Claude").

4. **Knowledge Base.ipynb** :

- This is the core of the project. It merges the data from the previous steps (topics, sentiment, and NER).
- It aggregates sentiment by both topic and LLM to find the performance of each model for each specific task.
- It calculates a "positive experience score" ($\text{positive} / (\text{positive} + \text{negative})$) for every LLM-topic pair (e.g., "Claude" + "Coding").

5. **LLM Recommender.ipynb** :

- Provides the final, interactive application.
- A user can type a natural language query (e.g., "I need help with debugging").
- The query is **semantically matched** to the most relevant topic ("Use Case: Coding & Development").
- The system then retrieves and ranks the best-performing LLMs for that specific task, based on the aggregated user sentiment from the knowledge base.



Releases

No releases published

Packages

No packages published

Languages

● Jupyter Notebook 100.0%

